Introduction

- The Schur fns. are characterized by the following two properties :—

  a) The transition matrix $M(s,m)$ that expresses the Schur fns. in terms of monomial symm. fns. m_λ is strictly upper triangular, that is,

  $$s_\lambda = m_\lambda + \sum_{\mu<\lambda} K_{\lambda\mu}\, m_\mu$$

  for suitable coeff. $K_{\lambda\mu}$ (the Kostka numbers).

  b) The s_λ are pairwise orthogonal (even orthonormal) relative to the scalar product defined by

  $$\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$$

- Here we study a class of symmetric fns. $P_\lambda(x; q,t)$ depending on two parameters q and t.

  They will be characterized by the same two properties (a) and (b), relative now to the scalar product defined by

  $$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_{i=1}^{l(\lambda)} \frac{1-q^{\lambda_i}}{1-t^{\lambda_i}} \qquad (1.5)$$

- When $q=t$, they reduce to the Schur functions s_λ.
- when $q=0$, they reduce to Hall-Littlewood fns. $P_\lambda(x;t)$
- when $q=t^\alpha$, in the limit $t\to 1$, they reduce to Jack's sym. fn.

Orthogonality

- Let q, t be independent indeterminates.
- Let $F = \mathbb{Q}(q,t)$ be the field of rational functions in q, t.
- Let $\Lambda_F = \Lambda \otimes_{\mathbb{Z}} F$ be the F-algebra of symmetric functions with coefficients in F.
- Define

$$(2.1)\qquad Z_\lambda(q,t) = z_\lambda \prod_{i=1}^{l(\lambda)} \frac{1-q^{\lambda_i}}{1-t^{\lambda_i}},$$

where $z_\lambda = \prod_{i \geq 1} i^{m_i}\, m_i!$.

Here $m_i = m_i(\lambda) = \#$ Parts of λ that are equal to i.

- The scalar product in (1.5) then takes the form

$$(2.2)\qquad \langle P_\lambda, P_\mu \rangle = \langle P_\lambda, P_\mu \rangle_{q,t} = \delta_{\lambda\mu}\, z_\lambda(q,t).$$

- We have $Z_\lambda(q^{-1}, t^{-1}) = (q^{-1}t)^{|\lambda|}\, Z_\lambda(q,t)$

$$\Rightarrow (2.3)\qquad \langle f, g \rangle_{q^{-1},t^{-1}} = (q^{-1}t)^n \langle f, g \rangle_{q,t}, \quad f, g \in \Lambda_F^n.$$

(homogeneous)

- For indeterminate a, let

$$(2.4)\qquad (a;q)_\infty = \prod_{r=0}^{\infty} (1 - aq^r)$$

which is regarded as a formal power series in a and q.

- Let $x = (x_1, x_2, \ldots)$ and $y = (y_1, y_2, \ldots)$ be two sequences of indeterminates, and define

(2.5) $$\Pi(x, y\,;\, q, t) = \prod_{i,j} \frac{(t x_i y_j\,;\, q)_\infty}{(x_i y_j\,;\, q)_\infty}.$$

Then we have

(2.6) $$\Pi(x, y\,;\, q, t) = \sum_\lambda z_\lambda(q, t)^{-1} P_\lambda(x) P_\lambda(y).$$

Proof :—

$$\Rightarrow \log \Pi(x, y\,;\, q, t) = \sum_{i,j} \sum_{r \geq 0} \log(1 - t x_i y_j q^r) - \log(1 - x_i y_j q^r)$$

$$= \sum_{i,j} \sum_{r \geq 0} \sum_{n \geq 1} \frac{(x_i y_j)^n}{n} q^{rn} (1 - t^n)$$

$$= \sum_{i,j} \sum_{n \geq 1} \frac{(x_i y_j)^n}{n} \frac{(1 - t^n)}{(1 - q^n)}$$

$$= \sum_{n \geq 1} \frac{1}{n} \frac{1 - t^n}{1 - q^n} P_n(x) P_n(y)$$

$$\Rightarrow \Pi(x, y\,;\, q, t) = \prod_{n \geq 1} \exp\left[\frac{1}{n} \frac{1 - t^n}{1 - q^n} P_n(x) P_n(y)\right]$$

$$= \prod_{n \geq 1} \sum_{m_n = 0}^{\infty} \left(\frac{1}{n} \frac{1 - t^n}{1 - q^n} P_n(x) P_n(y)\right)^{m_n} \frac{1}{m_n!}$$

$$= \prod_{n \geq 1} \sum_{m_n = 0}^{\infty} \frac{[P_n(x) P_n(y)]^{m_n}}{n^{m_n} m_n! \left(\frac{1 - q^n}{1 - t^n}\right)^{m_n}}$$

$$= \sum_\lambda z_\lambda(q, t)^{-1} P_\lambda(x) P_\lambda(y)$$



(2.7) For each $n \geq 0$, let $(u_\lambda), (v_\lambda)$ be F-bases of Λ^n_F, indexed by partitions of n. Then TFAE:

(a) $\langle u_\lambda, v_\mu \rangle = \delta_{\lambda\mu}$, for all λ, μ.

(b) $\sum_\lambda u_\lambda(x)\, v_\lambda(y) = \Pi(x, y; q, t)$

Proof :—

Let $P^*_\lambda = Z_\lambda(q,t)^{-1} P_\lambda$ so that $\langle P^*_\lambda, P_\mu \rangle = \delta_{\lambda\mu}$.

If $u_\lambda = \sum_\sigma a_{\lambda\sigma} P^*_\sigma$ and $v_\mu = \sum_\tau b_{\mu\tau} P_\tau$,

then $\langle u_\lambda, v_\mu \rangle = \sum_\sigma a_{\lambda\sigma} b_{\mu\sigma}$, so that

$$(a) \iff \sum_\sigma a_{\lambda\sigma} b_{\mu\sigma} = \delta_{\lambda\mu} \qquad (a')$$

Also, $\sum_\lambda u_\lambda(x)\, v_\lambda(y) = \sum_{\lambda,\sigma,\tau} a_{\lambda\sigma} b_{\lambda\tau} P^*_\sigma(x) P_\tau(y)$

So, $(b) \iff \sum_\sigma P^*_\sigma(x) P_\sigma(y) = \sum_{\lambda,\sigma,\tau} a_{\lambda\sigma} b_{\lambda\tau} P^*_\sigma(x) P_\tau(y)$

$$\iff \sum_{\sigma,\tau} \delta_{\sigma\tau} P^*_\sigma(x) P_\tau(y) = \sum_{\sigma,\tau} \Big(\sum_\lambda a_{\lambda\sigma} b_{\lambda\tau} \Big) P^*_\sigma(x) P_\tau(y)$$

$$\iff \sum_\lambda a_{\lambda\sigma} b_{\lambda\tau} = \delta_{\sigma\tau} \qquad (b')$$

$(a') \iff (b')$ since (a') is $AB^T = Id$.

while (b') is $A^T B = Id$.

∎

Let $g_n(x;q,t)$ be coeff. of y^n in the infinite product

$$(2.8)\qquad \prod_{i\geq 1} \frac{(tx_i y_1; q)_\infty}{(x_i y_1; q)_\infty} = \sum_{n\geq 0} g_n(x;q,t)\, y_1^n ,$$

and for any partition $\lambda = (\lambda_1, \lambda_2, \ldots)$, define

$$g_\lambda(x;q,t) = \prod_{i\geq 1} g_{\lambda_i}(x;q,t) .$$

Then we have :—

$$(2.9)\qquad g_n(x;q,t) = \sum_{|\lambda|=n} z_\lambda(q,t)^{-1} P_\lambda(x) ,$$

by setting $y_2 = y_3 = \cdots = 0$ in (2.6)

$$\Rightarrow \Pi(x,y;q,t) = \prod_{j\geq 1} \left(\sum_{n\geq 0} g_n(x;q,t)\, y_j^n \right)$$

$$\Rightarrow (2.10)\qquad \Pi(x,y;q,t) = \sum_{\lambda} g_\lambda(x;q,t)\, m_\lambda(y)$$

Then (2.7) implies

$$(2.11)\qquad \langle g_\lambda(x;q,t),\, m_\mu(x)\rangle = \delta_{\lambda\mu}$$

so that g_λ form a basis of Λ_F dual to the basis (m_λ).

Hence,

(2.12) The g_n $(n\geq 1)$ are algebraically independent over F, and $\Lambda_F = F[g_1, g_2, \ldots]$.

(2.13) Let $E : \Lambda_F \to \Lambda_F$ be an F-linear operator. Then TFAE:

(a) E is self-adjoint, that is, $\langle Ef, g\rangle = \langle f, Eg\rangle$, $f, g \in \Lambda_F$.

(b) $E_x\, \Pi(x, y; q, t) = E_y\, \Pi(x, y; q, t)$

(E_x: operation on x-variables; E_y: operation on y-variables.)

Proof :— For partitions λ, μ, let

$$e_{\lambda\mu} = \langle E m_\lambda, m_\mu \rangle .$$

Then (a) $\iff$ $e_{\lambda\mu} = e_{\mu\lambda}$ —— (c)

(2.11) implies $E m_\lambda = \sum_\mu e_{\lambda\mu} g_\mu$.

(2.10) implies $\Pi(x, y; q, t) = \sum_\lambda m_\lambda(x) g_\lambda(y) = \sum_\lambda m_\lambda(y) g_\lambda(x)$

Thus, (b) $\iff$ $\sum_{\lambda,\mu} e_{\lambda\mu}\, g_\mu(x) g_\lambda(y) = \sum_{\lambda,\mu} e_{\lambda\mu}\, g_\mu(y) g_\lambda(x)$

$\Updownarrow$

$e_{\lambda\mu} = e_{\mu\lambda}$ ▨

For $u, v \in F$ s.t. $v \neq \pm 1$, let $\omega_{u,v}$ be the F-algebra endomorphism of Λ_F, defined by

(2.14) $$\omega_{u,v}(p_r) = (-1)^{r-1} \frac{1-u^r}{1-v^r} p_r \quad , \quad r \geq 1$$

so that

(2.14′) $$\omega_{u,v}(P_\lambda) = \varepsilon_\lambda P_\lambda \prod_{i=1}^{\ell(\lambda)} \frac{1-u^{\lambda_i}}{1-v^{\lambda_i}} \quad , \text{ for partition } \lambda$$

where $\varepsilon_\lambda = (-1)^{|\lambda| - \ell(\lambda)}$

Clearly, $\omega_{v,u} = \omega_{u,v}^{-1}$ (if $u = \pm 1$)

and $\omega_{u,u}$ is the involution defined earlier by $\omega(e_r) = h_r$.
($\omega(p_n) = (-1)^{n-1} p_n$ was a consequence.)

From (2.9) and $e_n = \sum_{|\lambda|=n} \varepsilon_\lambda z_\lambda^{-1} p_\lambda$, it follows that

(2.15) $\omega_{q,t}(g_n(x;q,t)) = e_n(x)$,

the nth elementary symm. fn

The endomorphism $\omega_{u,v}$ is self-adjoint:

(2.16) $\langle \omega_{u,v} f, g \rangle = \langle f, \omega_{u,v} g \rangle$, $f, g \in \Lambda_F$.

By linearity it is enough to verify for $f = P_\lambda$, $g = P_\mu$ and in that case its immediate from definitions.

(2.17) $\langle \omega_{t,q} f, g \rangle = \langle \omega f, g \rangle_{q=t}$, $f, g \in \Lambda_F$.

we may take $f = P_\lambda$, $g = P_\mu$ in which case it is obvious.

(2.17) implies that (s_λ) and $(\omega_{t,q} s_{\lambda'})$ are dual bases of Λ_F

since $\omega s_{\lambda'} = s_\lambda$ or $\omega s_\lambda = s_{\lambda'}$

Finally, from (2.6) and $\prod_{i,j}(1+x_i y_j) = \sum_\lambda \varepsilon_\lambda z_\lambda^{-1} P_\lambda(x) P_\lambda(y)$,

it follows that

(2.18) $\omega_{q,t} \Pi(x,y;q,t) = \prod_{i,j}(1+x_i y_j)$

where $\omega_{q,t}$ acts on the symm. fns. in the x variables.

(2.19) The $g_\lambda(x;q,t)$ s.t. $l(\lambda)\leq n$ form an F-basis of $\Lambda_{n,F}$.

Proof :— $\Lambda_{n,F} = F[x_1,\dots,x_n]^{S_n}$ (P_λ no longer L.I.)

Proof. Suppose first that $q=t=0$. Then g_λ becomes h_λ in the notation of Chapter I, §2, and from Chapter I, §6 we have ↳ $h_n = \sum_{|\lambda|=n} z_\lambda^{-1} P_\lambda$

$$h_\mu(x) = \sum_\lambda K_{\lambda\mu} s_\lambda(x). \tag{1}$$

Now in the ring Λ_n the Schur function $s_\lambda(x)$ is zero if $l(\lambda)>n$, and the $s_\lambda(x)$ with $l(\lambda)\leqslant n$ from a $\mathbf{Z}$-basis of Λ_n. Since the matrix $(K_{\lambda\mu})$ is unitriangular, so is its principal submatrix $(K_{\lambda\mu})_{l(\lambda),\,l(\mu)\leqslant n}$. Hence it follows from (1) that the $h_\lambda(x)$ such that $l(\lambda)\leqslant n$ form a $\mathbf{Z}$-basis of Λ_n.

Now let q,t be arbitrary. Since the monomial symmetric functions $m_\mu(x)$ with $l(\mu)\leqslant n$ form a $\mathbf{Z}$-basis of Λ_n, we may write

$$g_\lambda(x;q,t) = \sum_\mu b_{\lambda\mu}(q,t) m_\mu(x) \tag{2}$$

with coefficients $b_{\lambda\mu}(q,t)\in F$. From above, the matrix $B(q,t) = (b_{\lambda\mu}(q,t))_{l(\lambda),\,l(\mu)\leqslant n}$ is well-defined and nonsingular when $q=t=0$. Hence its determinant, which is a rational function of q and t, cannot vanish identically. In other words, the matrix $B(q,t)$ is invertible over F, and hence (2) shows that the $g_\lambda(x;q,t)$ with $l(\lambda)\leqslant n$ form an F-basis of $\Lambda_{n,F}$, as required. |

We now define a (symmetric) scalar product on $\Lambda_{n,F}$ by requiring that the bases $(g_\lambda)_{l(\lambda)\leqslant n}$ and $(m_\lambda)_{l(\lambda)\leqslant n}$ be dual to each other:

$$\langle g_\lambda, m_\mu\rangle = \delta_{\lambda\mu} \tag{2.20}$$

whenever λ,μ are partitions of length $\leqslant n$. With the obvious modifications, (2.7) remains valid for this scalar product, since by virtue of (2.8) we have

$$\Pi(x,y;q,t) = \sum_{l(\lambda)\leqslant n} g_\lambda(x;q,t) m_\lambda(y) \tag{2.21}$$

when $x=(x_1,\dots,x_n)$ and $y=(y_1,\dots,y_n)$.

(*) $D_n^r : \Lambda_{n,F} \longrightarrow \Lambda_{n,F}$, $0 \le r \le n$

(F-linear endomorphisms.)

(3.1) $(T_{u,i}\, f)(x_1, \ldots, x_n) := f(x_1, \ldots, ux_i, \ldots, x_n)$, $u \in F$, $1 \le i \le n$, $f \in F[x_1, \ldots, x_n]$

(3.2) $D_n(X; q, t) = a_\delta(x)^{-1} \sum_{w \in S_n} (-1)^w x^{w\delta} \prod_{i=1}^{n} \left(1 + X\, t^{(w\delta)_i}\, T_{q,i}\right)$

$X \to$ indeterminate ; $(-1)^w \to$ sign of w

$\delta = (n-1, n-2, \ldots, 1, 0)$, $a_\delta(x) \to$ Vandermonde determinant, $(w\delta)_i$ is i^{th} component of $w\delta$.

$$a_\delta(x) = \det\left(x_i^{n-j}\right) = \prod_{1 \le i < j \le n} (x_i - x_j)$$

$$x^{w\delta} = \prod_{i=1}^{n} x_i^{(w\delta)_i}$$

(3.3) Let $D_n(X; q, t) = \sum_{r=0}^{n} D_n^r\, X^r$.

$r = 0 \Rightarrow D_n^0 = 1$ (Vandermonde over Vandermonde)

$r = 1 \Rightarrow$ (3.4) $D_n^1 = \sum_{i=1}^{n} A_i(x; t)\, T_{q,i}$

where $A_i(x; t) = a_\delta(x)^{-1} \sum_{w \in S_n} (-1)^w x^{w\delta}\, t^{(w\delta)_i}$

$= a_\delta(x)^{-1}\, T_{t,i}\, a_\delta(x)$

$\Rightarrow$ (3.5) $A_i(x; t) = \prod_{j \ne i} \dfrac{tx_i - x_j}{x_i - x_j}$

Pairs not involving i cancel. In pairs involving i if $j < i$ we multiply num and den by -1.

More generally, for any $r = 0, 1, \ldots, n$ we have

$$(3.4_r) \qquad D_n^r = \sum_{I} A_I(x;t) \prod_{i \in I} T_{q,i}$$

I: all r-element subsets of $\{1, 2, \ldots, n\}$.

where $A_I(x;t) = a_\delta(x)^{-1} \left(\prod_{i \in I} T_{t,i} \right) a_\delta(x)$

$$(3.5_r) \qquad = t^{r(r-1)/2} \prod_{\substack{i \in I \\ j \notin I}} \frac{tx_i - x_j}{x_i - x_j} .$$

$\binom{r}{2}$ pairs in I cancel as $\dfrac{tx_{i_1} - tx_{i_2}}{x_{i_1} - x_{i_2}}$

For $\nu \in \mathbb{N}^n$, $\lambda = (\lambda_1, \ldots, \lambda_n)$

$$D_n(X;q,t) x^\nu = a_\delta(x)^{-1} \sum_{w_1 \in S_n} (-1)^{w_1} \prod_{i=1}^{n} \left(1 + X t^{(w_1\delta)_i} q^{\nu_i}\right) x^{w_1\delta + \nu}$$

$$m_\lambda = \frac{1}{|S_n^\lambda|} \sum_{w_2 \in S_n} x^{w_2 \lambda}$$ where S_n^λ is subgroup of S_n that fixes λ.

$$\Rightarrow |S_n^\lambda| D_n(X;q,t) m_\lambda = a_\delta(x)^{-1} \sum_{w_1, w_2 \in S_n} (-1)^{w_1} \prod_{i=1}^{n} \left(1 + X t^{(w_1\delta)_i} q^{(w_2\lambda)_i}\right) x^{w_1\delta + w_2\lambda}$$

Let $w_2 = w_1 w$

$$\Rightarrow |S_n^\lambda| D_n(X;q,t) m_\lambda = a_\delta(x)^{-1} \sum_{w, w_1 \in S_n} (-1)^{w_1} \prod_{i=1}^{n} \left(1 + X t^{n-i} q^{(w\lambda)_i}\right) x^{w_1(\delta + w\lambda)}$$

$$= \sum_{w \in S_n} \prod_{i=1}^{n} \left(1 + X t^{n-i} q^{(w\lambda)_i}\right) \delta_{w\lambda}(x)$$

$$\Rightarrow D_n(X;q,t) m_\lambda = \sum_{\alpha} \prod_{i=1}^{n} \left(1 + X t^{n-i} q^{\alpha_i}\right) \delta_\alpha(x)$$

α: over all distinct derangements of λ.

Macdonald Polynomials II

Theorem

For each partition λ there is a unique symmetric function $P_\lambda(x;q,t) \in \Lambda_F$ such that:

(a) $P_\lambda = m_\lambda + \sum_{\mu<\lambda} u_{\lambda\mu} m_\mu$ with $u_{\lambda\mu} \in F$.

($<$: lexicographic ordering (total order); compatible with natural ordering (Partial): $\lambda \geq \mu \Leftrightarrow |\lambda| = |\mu|, \lambda_1 + \cdots + \lambda_r \geq \mu_1 + \cdots + \mu_r$ for $r \geq 1$)

(b) $\langle P_\lambda, P_\mu \rangle = 0$ if $\lambda \neq \mu$.

Remark: (a) and (b) overdetermine P_λ since for a diff. total order we get same P_λ by Gram–Schmidt.

→ Since $\langle , \rangle_{q,t}$ is scalar multiple (Proportional) of $\langle , \rangle_{q^{-1},t^{-1}}$, we have

$$P_\lambda(q^{-1},t^{-1}) = P_\lambda(q,t).$$

Proof :—

- Work initially with finite set of variables $x_1, \ldots, x_n$.
- Construct F-linear map $D : \Lambda_{n,F} \longrightarrow \Lambda_{n,F}$ ↳ $F[x_1,\ldots,x_n]^{S_n}$

having the properties:

① $Dm_\lambda = \sum_{\mu \leq \lambda} C_{\lambda\mu} m_\mu$ for each $\lambda : l(\lambda) \leq n$ (triangular)

② $\langle Df, g \rangle = \langle f, Dg \rangle$, $f, g \in \Lambda_F$ (self-adjoint)

③ $\lambda \neq \mu \Rightarrow C_{\lambda\lambda} \neq C_{\mu\mu}$ (distinct eigenvalues)

- Existence: Let $P_\lambda \in A_{n,F}$ be the eigenfunctions of D.

$$\Rightarrow DP_\lambda = c_{\lambda\lambda} P_\lambda.$$

Then (a) $\Leftrightarrow$ $c_{\lambda\lambda} u_{\lambda\nu} = \sum_{\mu:\, \nu \le \mu \le \lambda} u_{\lambda\mu} c_{\mu\nu}$

or $$DP_\lambda = \sum_{\mu \le \lambda} u_{\lambda\mu} D m_\mu = \sum_{\nu \le \mu \le \lambda} u_{\lambda\mu} c_{\mu\nu} m_\nu$$

$$\| $$

$$c_{\lambda\lambda} P_\lambda = \sum_{\nu \le \lambda} c_{\lambda\lambda} u_{\lambda\nu} m_\nu$$

$\Rightarrow \sum_{\mu \le \lambda} u_{\lambda\mu} m_\mu$ are eigenfunctions of D with eigenvalue $c_{\lambda\lambda}$.

$$D \sum_{\mu \le \lambda} u_{\lambda\mu} m_\mu = \sum_{\mu \le \lambda} u_{\lambda\mu} D m_\mu$$

as long as we have

$$= \sum_{\mu \le \lambda} u_{\lambda\mu} \sum_{\nu \le \mu} c_{\mu\nu} m_\nu$$

$$= \sum_{\mu \le \lambda} \sum_{\nu \le \mu} u_{\lambda\mu} c_{\mu\nu} m_\nu = c_{\lambda\lambda} \sum_{\nu \le \lambda} u_{\lambda\nu} m_\nu$$

Now $$c_{\lambda\lambda} u_{\lambda\nu} = \sum_{\substack{\mu \\ \nu \le \mu \le \lambda}} u_{\lambda\mu} c_{\mu\nu}$$

$$\Leftrightarrow \left(c_{\lambda\lambda} - c_{\nu\nu}\right) u_{\lambda\nu} = \sum_{\substack{\mu \\ \nu < \mu \le \lambda}} u_{\lambda\mu} c_{\mu\nu}$$

whenever $\nu < \lambda$

$c_{\lambda\lambda} - c_{\nu\nu} \neq 0$, $u_{\lambda\nu}$ is determined uniquely in terms of $u_{\lambda\mu} : \nu < \mu \le \lambda$

$\Rightarrow u_{\lambda\mu}$ in (a) are determined uniquely since $u_{\lambda\lambda} = 1$.

Orthogonality: $c_{\lambda\lambda}\langle P_\lambda, P_\mu\rangle = \langle c_{\lambda\lambda} P_\lambda, P_\mu\rangle$

$= \langle D P_\lambda, P_\mu\rangle = \langle P_\lambda, D P_\mu\rangle$

$= \langle P_\lambda, c_{\mu\mu} P_\mu\rangle = c_{\mu\mu}\langle P_\lambda, P_\mu\rangle$

Since, $c_{\lambda\lambda} \neq c_{\mu\mu}$ for $\lambda \neq \mu$, we have $\langle P_\lambda, P_\mu\rangle = 0$.

Finally, the coeff. $u_{\lambda\mu}$ can be computed by Gram-Schmidt involving scalar products $\langle m_\mu, m_\nu\rangle$ which are independent of n. $\Rightarrow P_\lambda$ are well-defined as elements of Λ_F.

Construction of D

$(T_{u,i} f)(x_1, \ldots, x_n) := f(x_1, \ldots, ux_i, \ldots, x_n)$, $u \in F$, $1 \le i \le n$, $f \in F[x_1, \ldots, x_n]$

(4) $D = a_\delta(x)^{-1} \sum_{\omega \in S_n} (-1)^\omega x^{\omega\delta} \sum_{i=1}^n t^{(\omega\delta)_i} T_{q,i}$

$= a_\delta(x)^{-1} \sum_{i=1}^n \sum_{\omega \in S_n} (-1)^\omega x^{\omega\delta} t^{(\omega\delta)_i} T_{q,i}$

$= a_\delta(x)^{-1} \sum_{i=1}^n \left(T_{t,i}\, a_\delta(x)\right) T_{q,i}$

$= \sum_{i=1}^n \left(\sum_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,i}$

Triangularity: let S_n^λ be subgroup of S_n that fixes λ.

$\Rightarrow m_\lambda = \frac{1}{|S_n^\lambda|} \sum_{\omega \in S_n} x^{\omega\lambda}$

$\Rightarrow |S_n^\lambda|\, D m_\lambda = a_\delta(x)^{-1} \sum_{\omega, \omega_1 \in S_n} (-1)^\omega \sum_{i=1}^n t^{(\omega\delta)_i} q^{(\omega_1\lambda)_i} x^{\omega_1\lambda + \omega\delta}$

Put $w_1 = w w_2$

$$\Rightarrow |S_n^\lambda| D m_\lambda = a_\delta(x)^{-1} \sum_{w, w_2 \in S_n} (-1)^w \underbrace{\sum_{i=1}^n t^{(w\delta)_i} q^{(w w_2 \lambda)_i}}\ x^{w(w_2\lambda+\delta)}$$

$$= a_\delta(x)^{-1} \sum_{w, w_2 \in S_n} \left(\sum_{i=1}^n t^{\delta_i} q^{(w_2\lambda)_i} \right) (-1)^w x^{w(w_2\lambda+\delta)}$$

$$= \sum_{w_2 \in S_n} \left(\sum_{i=1}^n q^{(w_2\lambda)_i} t^{n-i} \right) s_{w_2\lambda}$$

Let $u(\mu) = \sum_{i=1}^n q^{\mu_i} t^{n-i}$ for $\mu \in \mathbb{N}^n$.

Then $D m_\lambda = \sum_\mu u(\mu) s_\mu$

↑ over all distinct permutations of λ

$$s_\mu = \begin{cases} 0 & \text{if } \mu_i + n - i \text{ are not distinct} \\ \pm s_\nu & \text{for some partition } \nu \text{ s.t. } \mu + \delta = w(\nu + \delta) \end{cases}$$

↑ $(-1)^w$

$$\nu = w^{-1}\mu + w^{-1}\delta - \delta$$

$$\Rightarrow \nu < \lambda$$

$$\Rightarrow D m_\lambda = u(\lambda) s_\lambda + \sum_{\nu < \lambda} u(\nu) s_\nu$$

$$= \sum_{\mu \le \lambda} c_{\lambda\mu} m_\mu$$

with $c_{\lambda\lambda} = u(\lambda) = \sum_{i=1}^n q^{\lambda_i} t^{n-i}$.

↑ distinct eigenvalues

Self-adjoint : $\Pi^{-1} T_{q,i} \Pi = \prod_{j=1}^n \frac{1 - x_i y_j}{1 - t x_i y_j}$

is independent of q.

$\Rightarrow \pi^{-1} D_x \pi$ and $\pi^{-1} D_y \pi$ are both independent of q.

So, WLOG we assume $\underline{q = t}$.

$$\Rightarrow D = a_\delta(x)^{-1} \sum_{i=1}^{n} \left(T_{t,i}\, a_\delta(x)\right) T_{t,i}$$

$$\Rightarrow D s_\lambda = a_\delta(x)^{-1} \sum_{i=1}^{n} \left(T_{t,i}\, a_\delta(x)\right)\left(T_{t,i}\, s_\lambda(x)\right)$$

$$= a_\delta(x)^{-1} \sum_{i=1}^{n} T_{t,i}\left(a_\delta(x)\, s_\lambda(x)\right)$$

$$= a_\delta(x)^{-1} \sum_{i=1}^{n} T_{t,i}\left(\sum_{\omega \in S_n} (-1)^\omega x^{\omega(\lambda+\delta)}\right)$$

$$= a_\delta(x)^{-1}\left(\sum_{i=1}^{n} t^{\lambda_i + n - i}\right)\left(\sum_{\omega \in S_n} (-1)^\omega x^{\omega(\lambda+\delta)}\right)$$

$$= \left(\sum_{i=1}^{n} t^{\lambda_i + n - i}\right) s_\lambda$$

$$\Rightarrow \langle D s_\lambda, s_\mu \rangle = 0 \quad \text{for} \quad \lambda \neq \mu$$

$$\langle s_\lambda, D s_\mu \rangle = 0 \quad \text{for} \quad \lambda \neq \mu$$

and for $\lambda = \mu$, $\langle D s_\lambda, s_\lambda \rangle = \langle s_\lambda, D s_\lambda \rangle$

$\Rightarrow D$ is self-adjoint for $q = t$.

$\Rightarrow D_x \pi = D_y \pi \Rightarrow \pi^{-1} D_x \pi = \pi^{-1} D_y \pi$ (in general now)

$\Rightarrow D$ is self-adjoint (in general)

Examples

- $P_{(1^r)}(x; q, t) = m_{(1^r)} = e_r$.

For $\lambda \vdash 3$

$$P_{(1,1,1)} = m_{(1,1,1)} = e_3$$

$$P_{(2,1)} = m_{(2,1)} + u\, m_{(1,1,1)}$$

$$\Rightarrow u = -\frac{\langle m_{(2,1)}, m_{(1,1,1)}\rangle}{\langle m_{(1,1,1)}, m_{(1,1,1)}\rangle} = \frac{(t-1)(2qt+q+t+2)}{qt^2-1}$$

Illy, $P_3 = m_3 + \frac{(q^2+q+1)(t-1)}{q^2t-1} m_{21} + \frac{(q+1)(q^2+q+1)(t-1)^2}{(qt-1)(q^2t-1)} m_{111}$

$$m_{(2,1)} = P_1P_2 - P_3 = P_{21} - P_3$$

$$m_{(1,1,1)} = \frac{1}{3} P_{(3,0,0)} - \frac{1}{2} P_{(2,1,0)} + \frac{1}{6} P_{(1,1,1)}$$

$$\langle m_{(1,1,1)}, m_{(1,1,1)}\rangle = \frac{1}{9} z_3(q,t) + \frac{1}{4} z_{21}(q,t) + \frac{1}{36} z_{111}(q,t)$$

$$\langle m_{(2,1)}, m_{(1,1,1)}\rangle = -\frac{1}{2} z_{21}(q,t) - \frac{1}{3} z_3(q,t)$$

$$x_1^3 + x_2^3 + x_3^3 - 3x_1x_2x_3 = (x_1+x_2+x_3)(x_1^2+x_2^2+x_3^2-x_1x_2-x_2x_3-x_3x_1)$$

$$\Rightarrow P_3 - 3 m_{(1,1,1)} = P_1\left[P_2 - \frac{1}{2}(P_1^2 - P_2)\right] = \frac{3}{2} P_1P_2 - \frac{1}{2} P_1^3$$

$$\Rightarrow m_{(1,1,1)} = \frac{1}{3} P_3 - \frac{1}{2} P_1P_2 + \frac{1}{6} P_1^3$$

$$= \frac{1}{3} P_{(3,0,0)} - \frac{1}{2} P_{(2,1,0)} + \frac{1}{6} P_{(1,1,1)}$$

$$z_3(q,t) = 3\left(\frac{1-q^3}{1-t^3}\right), \quad z_{21}(q,t) = 2\left(\frac{1-q^2}{1-t^2}\right)\left(\frac{1-q}{1-t}\right)$$

$$z_{111}(q,t) = 6\left(\frac{1-q}{1-t}\right)^3$$

$$u = \frac{\frac{(1-q^2)(1-q)}{(1-t^2)(1-t)} + \left(\frac{1-q^3}{1-t^3}\right)}{\frac{1}{3}\left(\frac{1-q^3}{1-t^3}\right) + \frac{1}{2}\frac{(1-q^2)(1-q)}{(1-t^2)(1-t)} + \frac{1}{6}\frac{(1-q)^3}{(1-t)^3}}$$

$$u = 6\,\frac{(1-q^2)(1-q)(1-t^3)+(1-q^3)(1-t^2)(1-t)}{2(1-q^3)(1-t^2)(1-t)+3(1-q^2)(1-q)(1-t^3)+(1-q)^3(1+t)(1+t+t^2)}$$

$$= \frac{6(1-q)(1-t)\left[(1-q^2)(1+t+t^2)+(1-t^2)(1+q+q^2)\right]}{(1-q)\left[2(1+q+q^2)(1-t-t^2+t^3)+\right.}$$

$$= \frac{\cancel{6(1-q)}(1-t)\cancel{(1-qt)}(2+q+t+2qt)}{\cancel{b(1-q)}\cancel{(1-qt)}(1-qt^2)}$$

$$= \frac{(t-1)(2qt+q+t+2)}{(qt^2-1)}$$

Duality

(5) let $b_\lambda = b_\lambda(q,t) = \langle P_\lambda, P_\lambda\rangle_{q,t}^{-1} \in F$

(6) Define $Q_\lambda = b_\lambda P_\lambda$ so that $\langle P_\lambda, Q_\mu\rangle_{q,t} = \delta_{\lambda\mu}$.

i.e., (P_λ), (Q_λ) are dual bases of Λ_F.

Last time $\langle \omega_{q,t} f, g\rangle_{t,q} = \langle \omega f, g\rangle$ — (7)

$f, g \in \Lambda_F$

(8) **Theorem**

For all partitions λ we have

$$\omega_{q,t} P_\lambda(q,t) = Q_{\lambda'}(t,q)$$

or equivalently $\omega_{q,t} Q_\lambda(q,t) = P_{\lambda'}(t,q)$.

Proof :-

$$\boxed{f_\lambda(q,t) = (1-t)\sum_{i=1}^{\infty} q^{\lambda_i} t^{i-1}} \xleftarrow[\text{or Lemma}]{\text{claim (9)}} f_\lambda(q,t) = f_{\lambda'}(t,q).$$

$$f_\lambda(q,t) = (1-t)\sum_{i=1}^{\infty} q^{\lambda_i} t^{i-1}$$

$$= (1-t)\left[q^{\lambda_1} + q^{\lambda_2}t + \cdots + q^{\lambda_n}t^{n-1} + t^n + t^{n+1} + \cdots \infty\right]$$

$$= (1-t)\left[q^{\lambda_1} + q^{\lambda_2}t + \cdots + q^{\lambda_n}t^{n-1} + \frac{t^n}{1-t}\right]$$

$$= (1-t)\left[q^{\lambda_1} + q^{\lambda_2}t + \cdots + q^{\lambda_n}t^{n-1}\right] + t^n$$

$$= q^{\lambda_1} + (q^{\lambda_2} - q^{\lambda_1})t + (q^{\lambda_3} - q^{\lambda_2})t^2 + \cdots + (q^{\lambda_n} - q^{\lambda_{n-1}})t^{n-1} + (1-q^{\lambda_n})t^n$$

$$= (1-q)\left[\frac{1-q^{\lambda_n}}{1-q}t^n + q^{\lambda_n}\frac{1-q^{\lambda_{n-1}-\lambda_n}}{1-q}t^{n-1} + \cdots + q^{\lambda_2}\frac{1-q^{\lambda_1-\lambda_2}}{1-q}t + \frac{q^{\lambda_1}}{1-q}\right]$$

$\lambda_1 = \ell(\lambda') = m$, $n = \ell(\lambda) = \lambda'_1$, $\lambda'_i = \#\{j : \lambda_j \geq i\}$

$\lambda_i - \lambda_{i+1} = m_i(\lambda') = \#$ Parts of λ' equal to i.

$$= (1-q)\left[(1+q+\cdots+q^{\lambda_n-1})t^n + (q^{\lambda_n}+\cdots+q^{\lambda_{n-1}-1})t^{n-1} + \cdots + (q^{\lambda_2}+\cdots+q^{\lambda_1-1})t + q^{\lambda_1}+q^{\lambda_1+1}+\cdots\right]$$

$$= (1-q)\left[t^{n-\lambda'_1}q^0 + t^{\lambda'_2}q + \cdots + t^{\lambda'_m}q^{m-1} + q^m + q^{m+1} + \cdots\right]$$

$$= (1-q)\sum_{i=1}^{m} t^{\lambda'_i} q^{i-1} = f_{\lambda'}(t,q).$$

$$t^{-n} + t(1-t^{-1})\sum_{i=1}^{n} q^{\lambda_i} t^{-i}$$

$$= (1-t^{-1})\left[\sum_{i=1}^{n} q^{\lambda_i} t^{-(i-1)} + \frac{t^{-n}}{1-t^{-1}}\right] = f_\lambda(q,t^{-1})$$

$\downarrow t \to t^{-1}$

$$(1-t)\left[\sum_{i=1}^{n} q^{\lambda_i} t^{(i-1)} + \frac{t^n}{1-t}\right] = (1-t)\sum_{i=1}^{\infty} q^{\lambda_i} t^{i-1}$$

Let $E = E_{q,t} = t^{-n}\left(1 + (t-1)\, D_{q,t}\right)$

Then for any λ with $\ell(\lambda) \le n$,

$$E P_\lambda(q,t) = t^{-n}\left(1 + (t-1)\sum_{i=1}^{n} q^{\lambda_i} t^{n-i}\right) P_\lambda(q,t)$$

$$= f_\lambda(q,t^{-1})\, P_\lambda(q,t)$$

(*) $\omega_{q,t}\, E_{q,t}\, \omega^{-1}_{q,t} = E_{t^{-1},q^{-1}}$ (messy proof!)

Now, $E_{t^{-1},q^{-1}}\, \omega_{q,t} P_\lambda(q,t) = \omega_{q,t} E_{q,t} P_\lambda(q,t)$

$$= f_\lambda(q,t^{-1})\, \omega_{q,t} P_\lambda(q,t)$$

$$= f_{\lambda'}(t^{-1},q)\, \omega_{q,t} P_\lambda(q,t)$$

$\Rightarrow \omega_{q,t} P_\lambda(q,t)$ is eigenfunction of $E_{t^{-1},q^{-1}}$ with eigenvalue $f_{\lambda'}(t^{-1},q)$

$\Rightarrow$ it must be scalar multiple of $P_{\lambda'}(t^{-1},q^{-1}) = P_{\lambda'}(t,q)$

To show it is $Q_{\lambda'}(t,q)$ we must show

$$\langle \omega_{q,t} P_\lambda(q,t),\, P_{\lambda'}(t,q) \rangle_{t,q} = 1$$

By (7), equivalent to $\langle \omega P_\lambda(q,t),\, P_{\lambda'}(t,q) \rangle = 1$ (usual scalar prod.)

$$P_\lambda(q,t) = s_\lambda + \sum_{\mu<\lambda} a_{\lambda\mu}\, s_\mu$$

$$P_{\lambda'}(t,q) = s_{\lambda'} + \sum_{\nu^T<\lambda'} b_{\lambda\nu}\, s_{\nu'}$$

$$\omega s_\mu = s_{\mu'} \Rightarrow \omega P_\lambda(q,t) = s_{\lambda'} + \sum_{\mu' > \lambda'} a_{\lambda\mu} s_{\mu'}$$

$\Rightarrow \omega P_\lambda(q,t)$ and $P_{\lambda'}(t,q)$ have only $s_{\lambda'}$ in common.

$$\Rightarrow \langle \omega P_\lambda(q,t), P_{\lambda'}(t,q) \rangle = \langle s_{\lambda'}, s_{\lambda'} \rangle = 1.$$



(10) $$\sum_\lambda P_\lambda(x;q,t)\, Q_\lambda(y;q,t) = \Pi(x,y;q,t)$$ = dual bases.

↖ from (2.7) last time.

Last time $$\omega_{q,t}\, \Pi(x,y;q,t) = \prod_{i,j}(1+x_i y_j)$$ (2.18)

(11) $$\sum_\lambda P_\lambda(x;q,t)\, P_{\lambda'}(y;t,q) = \prod_{i,j}(1+x_i y_j)$$

↖ apply $\omega_{q,t}$ in y-variables

Then follows from Th (8) and last time (2.18)

when $q=t$

(10) becomes $$\sum_\lambda s_\lambda(x)\, s_\lambda(y) = \prod_{i,j}(1-x_i y_j)^{-1}$$

(11) becomes $$\sum_\lambda s_\lambda(x)\, s_{\lambda'}(y) = \prod_{i,j}(1+x_i y_j)$$

Specializations

$t=1$ $$D_{q,1} = \sum_{i=1}^{n} T_{q,i}$$

$\Rightarrow$ for $\lambda : \ell(\lambda) \le n$ we have

$$D_{q,1}\, m_\lambda = \left(\sum_{i=1}^{n} q^{\lambda_i}\right) m_\lambda$$

$\Rightarrow m_\lambda$ is eigenfunction of $D_{q,1}$

(12) $\Rightarrow P_\lambda(q,1) = m_\lambda$ for all partitions.

$\underline{q=1}$

From (8) (duality theorem)

$$\langle \omega_{q,t} P_\lambda(q,t), P_{\mu'}(t,q)\rangle_{t,q} = \delta_{\lambda\mu}$$

$$\Rightarrow \langle \omega P_\lambda(q,t), P_{\mu'}(t,q)\rangle = \delta_{\lambda\mu}$$

$$\Rightarrow \langle P_\lambda(q,t), \omega P_{\mu'}(t,q)\rangle = \delta_{\lambda\mu}$$

Put $t=1$

$$\Rightarrow \langle m_\lambda, \omega P_{\mu'}(1,q)\rangle = \delta_{\lambda\mu}$$

$$\Rightarrow \omega P_{\mu'}(1,q) = h_\mu \Rightarrow P_{\mu'}(1,q) = \omega h_\mu = e_\mu$$

replace q by t and μ' by λ

(13) $\Rightarrow P_\lambda(1,t) = e_{\lambda'}$ for all partitions λ.

(*) Compatibility with restriction homomorphisms

$$\tilde{E}_n = t^{-n} D_n - \sum_{i=1}^{n} t^{-i}$$

$$\tilde{E}_n m_\lambda = \sum_\alpha \left(\sum_{i=1}^{n} (q^{\alpha_i} - 1) t^{-i} \right) s_\alpha$$

since $D_n m_\lambda = \sum_\alpha \left(\sum_{i=1}^{n} q^{\alpha_i} t^{n-i} \right) s_\alpha$

$\Rightarrow \rho_{n,n-1} \circ \tilde{E}_n = \tilde{E}_{n-1} \circ \rho_{n,n-1}$, $\rho_{n,n-1} : \Lambda_{n,F} \to \Lambda_{n-1,F}$ defined by setting $x_n = 0$

Observe that $s_\alpha(x_1, \ldots, x_{n-1}, 0) = 0$ if $\alpha_n \neq 0$.

Properties of $P_\lambda(x; q, t)$

(14) $$P_{(r)} = \frac{(q;q)_r}{(t;q)_r}\, g_r .$$

Proof :-

g_r (by defn. of scalar product) is orthogonal to m_μ, for $\mu \neq (r)$

$\Rightarrow g_r$ is orthogonal to P_μ for $\mu \neq (r)$

$\Rightarrow g_r$ is a scalar multiple of $P_{(r)}$

Define $$(a;q)_m = \prod_{r=0}^{m-1} (1 - aq^r) = \frac{(a;q)_\infty}{(aq^m;q)_\infty}$$

and for any partition $\mu = (\mu_1, \mu_2, \ldots)$, let

$$(a;q)_\mu = \prod_{i \geq 1} (a;q)_{\mu_i}$$

Then we have $$g_n(x; q, t) = \sum_{|\mu| = n} \frac{(t;q)_\mu}{(q;q)_\mu}\, m_\mu(x)$$

This follows from

(15) $$\frac{(ty_1;q)_\infty}{(y_1;q)_\infty} = \sum_{m \geq 0} \frac{(t;q)_m}{(q;q)_m}\, y_1^m$$

by substituting $y_1 = x_i y_1$, $i \geq 1$

so that $$\prod_{i \geq 1} \frac{(tx_i y_1;q)_\infty}{(x_i y_1;q)_\infty} = \sum_{n \geq 0} \underbrace{\left(\sum_{|\mu| = n} \frac{(t;q)_\mu}{(q;q)_\mu}\, m_\mu(x) \right)}_{g_n(x;q,t)} y_1^n$$

(by defn. of g_n)

Now to prove (15), let

$$F(y_1) = \frac{(ty_1;q)_\infty}{(y_1;q)_\infty} = \prod_{n=0}^{\infty} \frac{1-ty_1q^n}{1-y_1q^n} = \sum_{m=0}^{\infty} A_m(q,t)\, y_1^m$$

$$\Rightarrow (1-y_1)F(y_1) = (1-ty_1) \prod_{n=1}^{\infty} \frac{1-ty_1q^n}{1-y_1q^n}$$

$$= (1-ty_1)\, F(qy_1)$$

Clearly $A_0 = F(0) = 1$, and

$$A_m - A_{m-1} = q^m A_m - t q^{m-1} A_{m-1}$$

$$\Rightarrow \quad A_m = \frac{1-tq^{m-1}}{1-q^m} A_{m-1}$$

$$\Rightarrow A_m = \frac{(1-tq^{m-1})(1-tq^{m-2})\cdots(1-t)}{(1-q^m)(1-q^{m-1})\cdots(1-q)} = \frac{(t;q)_m}{(q;q)_m}$$

(15) $P_\lambda(x_1,\ldots,x_n;q,t) = 0$ if $n < \ell(\lambda)$.

Proof :—

$\mu \le \lambda \Rightarrow \mu' \ge \lambda'$ so that

$\ell(\mu) = \mu_1' \ge \lambda_1' = \ell(\lambda) > n \Rightarrow m_\mu(x_1,\ldots,x_n) = 0$ for $\mu \ \lambda$

$\Rightarrow P_\lambda(x_1,\ldots,x_n) = 0$ if $n < \ell(\lambda)$

(16) Let λ be a partition with $\ell(\lambda) = n$. Then

$$P_\lambda(x_1,\ldots,x_n;q,t) = x_1x_2\cdots x_n\, P_\mu(x_1,\ldots,x_n;q,t)$$

where $\mu = (\lambda_1-1, \lambda_2-1, \ldots, \lambda_n-1)$.

Proof :—

$$T_{q,i}(x_1 x_2 \cdots x_n P_\mu) = q\, x_1 x_2 \cdots x_n\, T_{q,i}(P_\mu)$$

$$\Rightarrow D_n(x_1 \cdots x_n P_\mu) = q\, x_1 \cdots x_n\, D_n(P_\mu)$$

$$= q\left(\sum_{i=1}^{n} q^{\mu_i} t^{n-i}\right) x_1 \cdots x_n P_\mu$$

$$= \left(\sum_{i=1}^{n} q^{\lambda_i} t^{n-i}\right) x_1 \cdots x_n P_\mu$$

$\Rightarrow P_\lambda$ and $x_1 \cdots x_n P_\mu$ are proportional.

But they both have leading term x^λ

$\Rightarrow P_\lambda = x_1 \cdots x_n P_\mu$

(17) $$\omega_{q,t} E_{q,t} \omega_{q,t}^{-1} = E_{t^{-1}, q^{-1}}$$

Proof (outline) :-

(a) For $r \geq 0$ and $x = (x_1, \ldots, x_n)$

$$(t-1)\sum_{i=1}^{n} A_i(x;t)\, x_i^r = t^n\, g_r(x; 0, t^{-1}) - \delta_{0r}.$$

$$\hookrightarrow A_i(x;t) = \prod_{j \neq i} \frac{t x_i - x_j}{x_i - x_j}.$$

Express the product $\prod_{i=1}^{n} \frac{tX - x_i}{X - x_i}$ as a sum of partial fractions.

(b) $\Pi = \Pi(x, y; q, t)$ for $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$

and let $\Pi_0 = \omega_{q,t} \Pi = \prod_{i,j} (1 + x_i y_j)$

($\omega_{q,t}$ acts on x-variables)

Then :

$$\Pi^{-1} T_{q,i} \Pi = \sum_{r \geq 0} g_r(y; 0, t^{-1})\, t^r x_i^r \,,$$

$$\Pi_0^{-1} T_{q,i} \Pi_0 = \sum_{r \geq 0} (-1)^r g_r(y; 0, q)\, x_i^r \,.$$

(c) $$\Pi^{-1} E_{q,t} \Pi = \sum_{r \geq 0} g_r(x; 0, t^{-1}) g_r(y; 0, t^{-1})\, t^r \,,$$

$$\Pi_0^{-1} E_{q,t} \Pi_0 = \sum_{r \geq 0} (-1)^r g_r(x; 0, t^{-1}) g_r(y; 0, q)$$

where $E_{q,t}$ acts in the x-variables.

(d) $$\omega_{q,t}\, g_r(x; 0, t^{-1}) = (-t)^{-r} g_r(x; 0, q)$$

(e) $$\omega_{q,t} (\Pi^{-1} E_{q,t} \Pi) = \Pi_0^{-1} E_{t^{-1}, q^{-1}} \Pi_0$$

$$\Rightarrow \quad \omega_{q,t} E_{q,t} \omega_{q,t}^{-1} = E_{t^{-1}, q^{-1}} \,.$$

$$\prod_{i\geq 1}(1-x_i t)^{-1} = H(t) = \sum_{\lambda} z_\lambda^{-1} p_\lambda t^{|\lambda|}$$

$$\prod_{i\geq 1}(1+x_i t) = E(t) = \sum_{\lambda} \varepsilon_\lambda z_\lambda^{-1} p_\lambda t^{|\lambda|}$$

$$P(t) = \frac{H'(t)}{H(t)} \Rightarrow \ln' H(t) = \sum_{r\geq 1} p_r t^{r-1}$$

$$\Rightarrow \ln H(t) = \sum_{r\geq 1} \frac{p_r}{r} t^r$$

$$\Rightarrow H(t) = \exp \sum_{r\geq 1} p_r \frac{t^r}{r} = \prod_{r\geq 1} \exp\left(p_r \frac{t^r}{r}\right)$$

$$= \prod_{r\geq 1} \sum_{m_r=0}^{\infty} \frac{(p_r t^r)^{m_r}}{r^{m_r} \cdot m_r!}$$

$$= \sum_{\lambda} z_\lambda^{-1} p_\lambda t^{|\lambda|}$$